COMP2007: Operating Systems & Concurrency
Week 9 – 12:00pm Wednesday – 20 November 2024

valid for 75 minutes from 11:45am
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Figure: Attendance Monitoring

# Operating Systems and Concurrency

## File Systems 1

## COMP2007

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2024

# Goals for Today

## Overview

- **Rotational** and **solid state** drives
- **Access times** for hard drives
- **Disk scheduling**

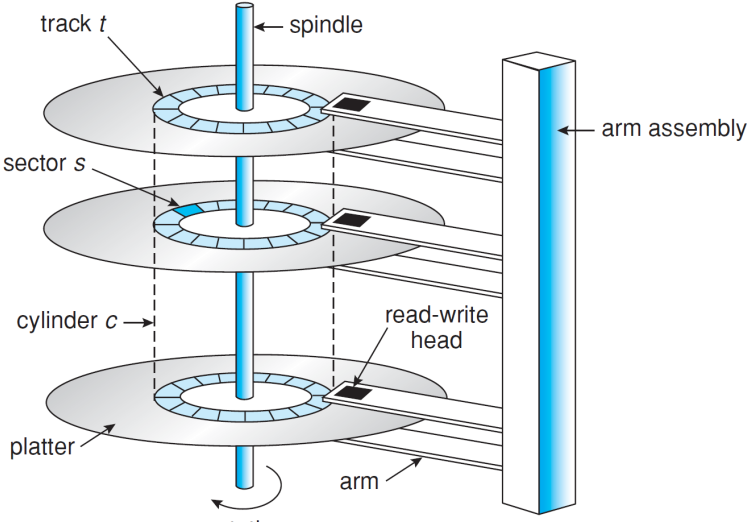
# Rotational Hard Drives

## Construction

- Rotational hard drives are made of aluminium/glass **platters** covered with **magnetisable material**
  - **Read/write heads** fly just above the surface (0.2 – 0.07mm) and are connected to a single **disk arm** controlled by a single **actuator**
  - **Data** is stored on both sides
  - Common **diameters** range from 1.8 to 3.5 inches
  - They **rotate** at a **constant speed** (speed near the spindle is less than on the outside)
- A **disk controller** abstracts the low level interface
- **Rotational hard drives** are approx. **4 orders of magnitude slower than main memory**

# Rotational Hard Drives

## Construction



# Rotational Hard Drives

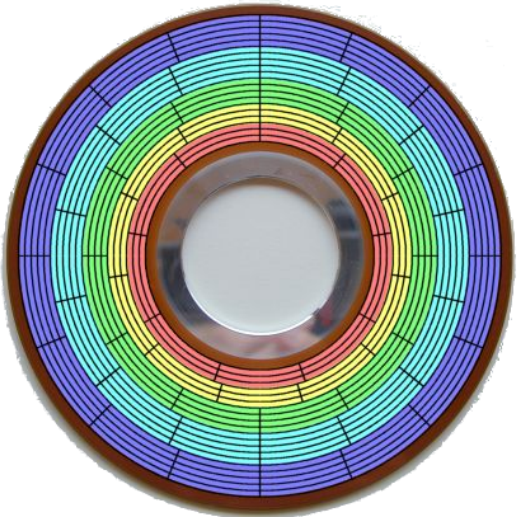
## Low Level Format

- Disks are organised in:
  - **Tracks**: a concentric circle on a single platter side
  - **Sectors**: segments of a track (usually 512B or 4KB in size)
  - **Cylinders**: all tracks in the same position relative to the spindle
- **Sectors** usually have an **equal number of bytes** in them (**preamble**, **data**, **error correcting code - ECC**)
- The **number of sectors increases** from the inner side of the disk to the outside

Figure: Disk Sector

# Rotational Hard Drives

## Organisation of Rotational Drives



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# Rotational Hard Drives

## Organisation

- **Cylinder skew** is an **offset** that is added to sectors in adjacent tracks to account for the seek time
- In the past, consecutive **disk sectors were interleaved** to account for transfer time
- **Disk capacity** is reduced due to **low level formatting** (preamble, ECC, etc.)

# Rotational Hard Drives

## Access Times

- **Access time** = seek time + rotational delay + transfer time
  - **Seek time**: time needed to move the arm to the cylinder (dominant)
  - **Rotational latency**: time before the sector appears under the head
  - **Transfer time**: time to transfer the data

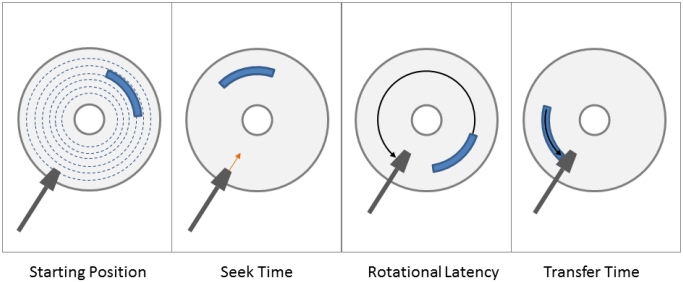
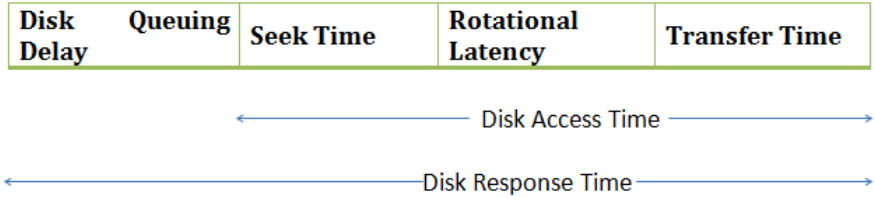



Figure: Access time to Disk (Source: www.studiodaily.com/)

# Rotational Hard Drives

## Access Times

- Multiple requests may be happening at the same time (concurrently). Thus, access time may be increased by a **queueing time**
- Dominance of seek time leaves room for **optimisation** by carefully considering the order of read operations



# Rotational Hard Drives

## Access Times

- The **estimated seek time** $T_s$ to move the arm from one track to another is approximated by:

$$T_s = n \times m + s \tag{1}$$

- In which:
  - $n$ the **number of tracks** to be crossed
  - $m$ the **crossing time per track**
  - $s$ any **additional startup delay**

# Rotational Hard Drives

## Access Times

- Assume a disk that **rotates at 3600rpm** (common rotation speeds are between 3600 and 15000rpm)
  - One rotation takes approx. 16.7ms ($\frac{60000}{3600}$)
  - The average **rotational latency** ($T_r$) is half a rotation on average ($\frac{16.7}{2} \approx 8.3ms$)
- Let $b$ denote the **number of bytes transferred**, $N$ the **number of bytes per track**, and $rpm$ the **rotation speed** in revolutions per minute
- The **transfer time** $T_t$ is given by:
  - $N$ bytes take 1 revolution (16.7ms)
  - $b$ contiguous bytes takes $\frac{b}{N}$ revolutions

$$T_t = \frac{b}{N} \times ms\ per\ revolution \tag{2}$$

# Rotational Hard Drives

## Access Times: Example

- To read a file of **size 256 sectors** with:
  - $T_s$ = 20 ms (average seek time)
  - 32 sectors/track
- If the file is stored **contiguously**:
  - The first track: $20 + \frac{16.7}{2} + 16.7 = 45ms$ (seek + rotational delay + transfer time)
  - The remaining tracks (assuming no cylinder skew and negligible seeks time between neighbouring tracks): $8.3 + 16.7 = 25ms$ (rotational delay + transfer time)
- The total time is then $45 + 7 \times 25 = 220ms$ = 0.22s

# Rotational Hard Drives

## Access Times: Example

- In case the access is not sequential but at **random for the sectors**, we get:
  - Time per sector = $T_s + T_r + T_t = 20 + \frac{16.7}{2} + \frac{16.7}{32} = 28.8ms$
  - Total time for 256 sectors = $256 \times 28.8ms = 7.37s$
- Observation: **sectors** must be **positioned carefully** and avoid **disk fragmentation**

# Disk Scheduling

## Concepts

- The OS/hardware must:
  - **Position/organise** files and sectors strategically
  - Optimise **disk requests** to **minimise ovehead** from seek time and rotational delays
- **I/O requests happen over time** and go through a **system calls** and are **queued**:
  - They are kept in a **table of requested sectors per cylinder**
  - This allows the operating system to **intercept** and **re-sequence** them

# Disk Scheduling

## Concepts

- **Disk scheduling algorithms** determine the order in which disk requests are processed to **minimise overhead**
  - They commonly use **heuristic approaches**
  - That is, **none** of the algorithms discussed here are **optimal algorithms**
- Assume a disk with 36 cylinders, numbered 1 to 36

# Disk Scheduling

## First-Come, First-Served

- **First come first served**: process the requests in the order that they arrive
- Consider the following sequence of disk requests (cylinder locations):

  `11 1 36 16 34 9 12`
- In the order of arrival (FCFS) the total length is:

  `|11−1|+|1−36|+|36−16|+|16−34|+|34−9|+|9−12|=111`

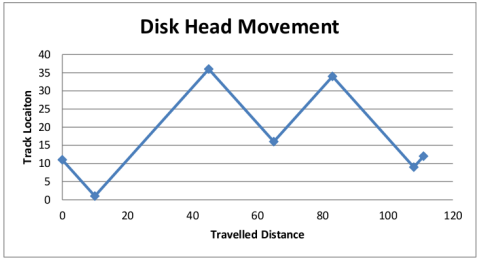



Figure: Head movement for FCFS

# Disk Scheduling

## Shortest Seek Time First

- **Shortest seek time first** selects the request that is **closest to the current head position**
- In the order "**shortest seek time first**" (SSTF) we gain approx. **50%**:

```
11 1 36 16 34 9 12:
|11−12|+|12−9|+|9−16|+|16−1|+|1−34|+|34−36|=61
```

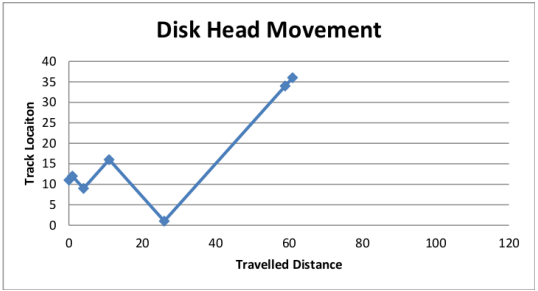



Figure: Head movement for shortest seek time

# Disk Scheduling

## Shortest Seek Time First

### Shortest Seek Time First

Any shortcomings or problems?

# Disk Scheduling

## Shortest Seek Time First

### Shortest Seek Time First

Any shortcomings or problems?

- The **arm stays in the middle of the disk** in case of heavy load (edge cylinders are poorly served)
- Continuously arriving requests for the same location could **starve** other regions

# Disk Scheduling

## SCAN

### Starvation

How would you solve it?

# Disk Scheduling

## SCAN

### Starvation

How would you solve it?

- "Lift algorithm, **SCAN**": **keep moving in the same direction** until end is reached (start upwards):
  - It continues in the current direction, **servicing all pending requests** as it passes over them
  - When it gets to the **last cylinder**, it **reverses direction** and services all the pending requests (until it reaches the first cylinder)
  - The **upper limit** on the "waiting time" is $2 \times$ number of cylinders, i.e. **no starvation occurs**
- Look-SCAN moves to the cylinder containing **the first/last request** (as opposed to the first/last cylinder on the disk with SCAN)

# Disk Scheduling

## SCAN

- "Lift algorithm, SCAN":

```
11 1 36 16 34 9 12:
|11-12|+|12-16|+|16-34|+|34-36|+|36-9|+|9-1|=60
```

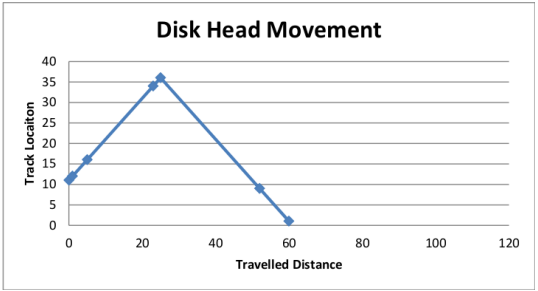



Figure: Head movement for SCAN

# Disk Scheduling

## SCAN

### SCAN Algorithm

Any shortcomings or problems?

# Disk Scheduling

## SCAN

### SCAN Algorithm

Any shortcomings or problems?

- The **middle cylinders are favoured** if the disk is heavily used (max. wait time is N tracks, 2N for the cylinders on the edge)
- Seeks are **cylinder by cylinder** and one cylinder contains multiple track, the arm can **stick to a cylinder**

# Disk Scheduling

## C-SCAN

### Fairness

How would you solve it?

# Disk Scheduling

## C-SCAN

### Fairness

How would you solve it?

- Once the outer/inner side of the disk is reached, the **requests at the other end of the disk have been waiting longest**
- SCAN can be improved by using a **circular scan** approach ⇒ C-SCAN
  - The disk arm moves in **one direction** servicing requests until the **last cylinder** is reached
  - It **reverses direction** but **does not service requests** when returning
  - Once it gets back to the **first cylinder** it reverses direction and **services requests**
  - It is **fairer** and **equalises response times** across a disk
- The C-SCAN algorithm (for `11 1 36 16 34 9 12`):
  `|11-12|+|12-16|+|16-34|+|34-36|+|36-1|+|1-9|=68`

# Disk Scheduling

## Other SCAN variations: LOOK-SCAN, N-step-SCAN

- N-step-SCAN: **divide the queue** into segments of *N* requests which are handled using SCAN
- F-SCAN: **use two queues**, while one is handled with SCAN, the other one is filled (swapped over when all are processed)

# Disk Scheduling

## Observations

- Look-SCAN and variations are reasonable choices for the algorithms
- Performance of the **algorithms is dependent on the requests/load of the disk**
  - One request at a time $\Rightarrow$ FCFS will perform equally well as any other algorithm
- **Optimal algorithms** are difficult to achieve if requests arrive over time (they need perfect knowledge of information)

# Rotational Hard Drives

## Driver Caching

- For most current drives, **the time required to seek** a new cylinder is **more than the rotational time** (remember **pre-paging** in this context!)
- It makes sense, therefore, to **read more sectors than actually required**
  - **Read** sectors during the **rotational delay** (i.e. that accidentally pass by)
  - **Modern controllers read multiple sectors** when asked for the data from one sector: track-at-a-time caching

# SSD drives

## Architecture

- Solid State Drives (SSDs):
  - Have **no delays** due to **no moving parts**, store data using **single level** (SLC), **multiple level** (MLC), **triple level** (TLC) electrical circuits, and suffer from **wear out** and **disturbance**
  - Are organised into **banks**, **blocks**, **pages** and have some **volatile cache memory** (buffering, mapping tables)
  - Often use multiple **banks in parallel** to improve performance

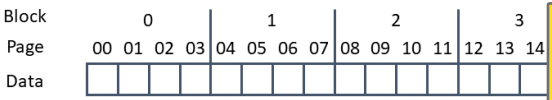



Figure: Layout of an SSD

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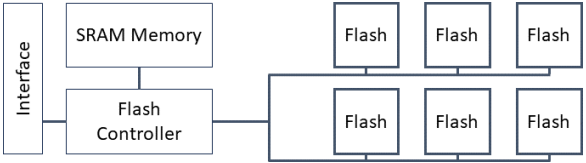


Figure: Layout of an SSD

# SSD drives

## Reads/Writes

- The **Flash Translation Layer** that maps **logical blocks** onto **physical pages**
- The following operations are supported:
  - Read: uniformly fast random access to any **page** to any location (10s of microseconds)
  - Erase: entire **blocks** containing multiple pages (milliseconds magnitude)
  - Program: write a **page** (100s of microseconds, block must be erased first)

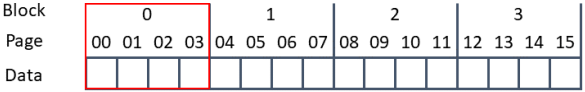


Figure: SSD Write Operation

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| Block | 0 | | | |
|---|---|---|---|---|
| Page | 00 | 01 | 10 | 11 |
| Data | 00101001 | 10100101 | 11101011 | 00001010 |

Figure: SSD Write Operation

# SSD drives

## Reads/Writes

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| Block | 0 | | | |
|---|---|---|---|---|
| Page | 00 | 01 | 10 | 11 |
| Data | 11111111 | 11111111 | 11111111 | 11111111 |

Figure: SSD Write Operation

# SSD drives

## Reads/Writes

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| Block | 0 | | | |
|---|---|---|---|---|
| Page | 00 | 01 | 10 | 11 |
| Data | 10101010 | 11111111 | 11111111 | 11111111 |

Figure: SSD Write Operation

# SSD drives

## Direct Mapping

- If using direct mapping, **logical pages** (seen by the OS) are **directly mapped** on **physical pages**
  1. Read the entire block for the given page
  2. Erase the entire block
  3. Write the new page and remaining old pages back
- Write performance is bad (**write amplification**) and **wear is increased** (some blocks are used more than others) $\Rightarrow$ a different **log structured approach** is needed

| Device | Read ($\mu$s) | Program ($\mu$s) | Erase ($\mu$s) |
|---|---|---|---|
| SLC | 25 | 200-300 | 1500-2000 |
| MLC | 50 | 600-900 | ~3000 |
| TLC | ~75 | ~900-1350 | ~4500 |

Figure: SSD Performance (from Arpaci-Dusseau)

# SSD drives

## Direct Mapping

| Device | Random Reads (MB/s) | Random Writes (MB/s) | Sequential Reads (MB/s) | Sequential Writes (MB/s) |
|---|---|---|---|---|
| Samsung 840 Pro SSD | 103 | 287 | 421 | 384 |
| Seagate 600 SSD | 84 | 252 | 424 | 374 |
| Intel SSD 335 SSD | 39 | 222 | 344 | 354 |
| Seagate Savvio 15K.3 HDD | 2 | 2 | 223 | 223 |

Figure: SSD / HDD Comparison (from Arpaci-Dusseau)

# Test Your Understanding

## Problem (from Tanenbaum)

### Disk Access Times

Disk requests come in to the disk driver for cylinders 10, 22, 20, 2, 40, 6 and 38, in that order.

- A seek takes 6ms per cylinder.
- How much seek time is needed for: FCFS, SSTF and Look-SCAN (initially moving upward)
- In all cases, the arm is initially at cylinder 20.

# Summary

## Take-Home Message

- Construction and organisation of **rotational hard drives**
- **Access times** of rotational hard drives
- **Disk scheduling** & caching
- **Solid state drives**

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